蔡栋琪

南京大学（苏州校区） 智能科学与技术学院

邮箱 cdq@nju.edu.cn

个人主页：http://www.caidongqi.com/

研究方向

具身智能系统，端侧大模型，软硬件协同优化

工作/教育经历

03/2026 至今 **准聘助理教授**（特聘研究员，博导），南京大学

08/2024 - 08/2025 **联合培养博士生**，剑桥大学 St John's College

- 合作导师：Nicholas D. Lane

09/2021 - 11/2025 **博士研究生** 计算机科学与技术专业，北京邮电大学

- 导师：王尚广、徐梦炜
- 合作导师： Felix Xiaozhu Lin

09/2017 - 06/2021 **学士** 通信工程专业，北京邮电大学

实习经历

07/2021 - 12/2021. **算法实习生**, 微众银行

- 企业导师：范力欣
- 部门负责人：杨强

奖项与荣誉

- 首届中国科协青年人才托举工程（博士生特别项目），2024
- Best Poster Award, MobiUK, 2025
- MobiSys Rising Star，SigMobile，2025
- 国家奖学金，教育部，2023/2024（连续两年）
- 国家留学基金委员会（CSC）奖学金，2024
- Travel Grant, NeurIPS'24/EuroSys'24/MobiSys'24/ATC'24/MobiSys'25/MobiUK'25
- 北京邮电大学优秀研究生，2023
- 网络与交换技术国家重点实验室优秀研究生，2022/2023

部分期刊论文 (* = 同等贡献; # = 通讯作者)

[J6] "FwdLLM+: Accelerating Forward-only FedLLM with Low-rank Perturbations"

Chen Peng, Mengwei Xu, Zhenyan Lu, Wei Liu, Shangguang Wang, Nicholas D Lane, Qibo Sun, **Dongqi Cai**[#], *IEEE Transactions on Mobile Computing (**TMC, CCF-A**),* 2026.

[J5] "Auditing unauthorized training data from AI generated content using information isotopes"

Tao Qi, Jinhua Yin, **Dongqi Cai**, Yueqi Xie, Huili Wang, Zhiyang Hu, Peiru Yang, Guoshun Nan, Zhili Zhou, Chuhan Wu, Lingjuan Lyu, Shangguang Wang, Yongfeng Huang, Nicholas D Lane, ***Nature Communications (Nature 子刊)***, 2026.

[J4] "Ubiquitous Memory Augmentation via Mobile Multimodal Embedding System"

Dongqi Cai, Shangguang Wang, Chen Peng, Zeling Zhang, Zhenyan Lu, Tao Qi, Nicholas D. Lane, Mengwei Xu, ***Nature Communications (Nature 子刊)***, 2025.

[J3] "面向微控制单元的高效语音隐私保护编码器"

蔡栋琪, 王尚广, 张泽凌, 马骁, 徐梦炜, ***电子学报 (CCF-A 中文期刊)***, 2025.

[J2] "Resource-efficient Algorithms and Systems of Foundation Models: A Survey"

Mengwei Xu* (指导老师), **Dongqi Cai***, Wangsong Yin*, Shangguang Wang, Xin Jin, Xuanzhe Liu, accepted in *ACM Computing Surveys (**ACM CSUR, 中科院一区**)*, 2024.

[J1] "Accelerating Vertical Federated Learning"

Dongqi Cai, Tao Fan, Yan Kang, Lixin Fan, Mengwei XU, Shangguang Wang, Qiang Yang, e in *IEEE Transactions on Big Data (**IEEE TBD， 中科院二区**)*, 2024.

部分会议论文 (* = 同等贡献; # = 通讯作者)

[C10] "MobiEdit: Resource-efficient Knowledge Editing for Personalized On-device LLMs"

Zhenyan Lu, Daliang Xu, **Dongqi Cai**, Zexi Li, Wei Liu, Fangming Liu, Shangguang Wang, Mengwei Xu, in the *Fourteenth International Conference on Learning Representations (**ICLR, CCF-A**)*, 2026. **MobiUK 2025 Best Poster Award**

[C10] "Editing as Unlearning: Are Knowledge Editing Methods Strong Baselines for Large Language Model Unlearning?"

Zexi Li, Xiangzhu Wang, William F. Shen, Meghdad Kurmanji, Xinchi Qiu, **Dongqi Cai**, Chao Wu, Nicholas D. Lane, in the 40th Annual AAAI Conference on Artificial Intelligence *(**AAAI, CCF-A**),* 2026

[C9] "ShortcutsBench: A Large-Scale Real-world Benchmark for API-based Agents"

Haiyang Shen, Yue Li, Desong Meng, **Dongqi Cai**, Sheng Qi, Li Zhang, Mengwei Xu, Yun Ma, in the *Thirteenth International Conference on Learning Representations (**ICLR, CCF-A**)*, 2025.

[C8] "Photon: Federated Pre-training of Large Language Models"

Lorenzo Sani, Alex Iacob, Zeyu Cao, Royson Lee, Bill Marino, Yan Gao, Wanru Zhao, **Dongqi Cai**, Zexi Li, Xinchi Qiu, Nicholas Donald Lane, in the *Eighth Annual Conference on Machine Learning and Systems (**MLSys**)*, 2025.

[C7] "DEPT: Decoupled Embeddings for Pre-training Language Models"

Alex Iacob, Lorenzo Sani, Meghdad Kurmanji, William F. Shen, Xinchi Qiu, **Dongqi Cai**, Yan Gao, Nicholas Donald Lane, in the *Thirteenth International Conference on Learning Representations (**ICLR, CCF-A, [Oral, top 1.8%]**)*, 2025.

[C6] "Demystifying Small Language Models for Edge Deployment"

Zhenyan Lu, Xiang Li, **Dongqi Cai**, Rongjie Yi, Fangming Liu, Wei Liu, Jian Luan, Xiwen Zhang, Nicholas D. Lane, Mengwei Xu, in *the 63rd Annual Meeting of the Association for Computational Linguistics (**ACL, CCF-A**)*, 2025.

[C5] "Mobile Foundation Model as Firmware"

Jinliang Yuan*, Chen Yang*, **Dongqi Cai***, Shihe Wang, Xin Yuan, Zeling Zhang, Xiang Li, Dingge Zhang, Hanzi Mei, Xianqing Jia, Shangguang Wang, Mengwei Xu, in *Proc. ACM Int. Conf. Mobile Computing and Networking (**MobiCom, CCF-A**)*, 2024.

[C4] "FwdLLM: Efficient Federated Finetuning of Large Language Models with Perturbed Inferences"

Mengwei Xu (博士导师), **Dongqi Cai#**, Yaozong Wu, Xiang Li, Shangguang Wang, in *USENIX Annual Technical Conference (**USENIX ATC, CCF-A**),* 2024.

[C3] "Efficient Federated Learning for Modern NLP"

Dongqi Cai, Yaozong Wu, Shangguang Wang, Felix Xiaozhu Lin, Mengwei Xu, in *Proc. ACM Int. Conf. Mobile Computing and Networking (**MobiCom, CCF-A**)*, 2023.

[C2] "Federated Few-shot Learning for Mobile NLP"

Dongqi Cai, Shangguang Wang, Yaozong Wu, Felix Xiaozhu Lin, Mengwei Xu, in *Proc. ACM Int. Conf. Mobile Computing and Networking (**MobiCom, CCF-A**)*, 2023.

[C1] "SILENCE: Protecting privacy in offloaded speech understanding on wimpy devices"

Dongqi Cai, Shangguang Wang, Zeling Zhang, Felix Xiaozhu Lin, Mengwei Xu, in *the Annual Conference on Neural Information Processing Systems (**NeurIPS, CCF-A**),* 2024.

Workshop 论文(* = 同等贡献)

[W4] "Large Language Models on Mobile Devices: Measurements, Analysis, and Insights"

Xiang Li, Zhenyan Lu, **Dongqi Cai**, Xiao Ma, Mengwei Xu, in *Proceedings of the Workshop on Edge and Mobile Foundation Models (EdgeFM), co-located with ACM International Conference on Mobile Systems, Applications, and Services (**MobiSys, CCF-B**),* 2024.

[W3] "FedRDMA: Communication-Efficient Cross-Silo Federated LLM via Chunked RDMA Transmission"

Zeling Zhang*, **Dongqi Cai***, Yiran Zhang, Mengwei Xu, Shangguang Wang, Ao Zhou, in *Proceedings of the 4rd Workshop on Machine Learning and Systems (EuroMLSys), co-located with European Conference on Computer Systems (**EuroSys, CCF-A**),* 2024.

[W2] "Towards Practical Few-shot Federated NLP"

Dongqi Cai, Yaozong Wu, Haitao Yuan, Shangguang Wang, Felix Xiaozhu Lin, Mengwei Xu, in *Proceedings of the 3rd Workshop on Machine Learning and Systems (EuroMLSys), co-*

*located with European Conference on Computer Systems (**EuroSys, CCF-A**),* 2023.

[W1] "Towards ubiquitous learning: A first measurement of on-device training performance"

Dongqi Cai, Qipeng Wang, Yuanqiang Liu, Yunxin Liu, Shangguang Wang, Mengwei Xu, in *Proceedings of the 5th International Workshop on Embedded and Mobile Deep Learning* (*EMDL*), *co-located with ACM International Conference on Mobile Systems, Applications, and Services (**MobiSys, CCF-B**),* 2021.

专利

[P1] 纵向联邦学习建模优化方法、设备、介质及程序产品。**蔡栋琪**，范力欣，杨强

[P2] 一种面向预训练模型的联邦学习方法、装置及系统。徐梦炜，**蔡栋琪**，周傲，马骁，王尚广

[P3] 面向自然语言模型的联邦小样本学习方法、系统及设备。徐梦炜，**蔡栋琪**，周傲，马骁，王尚广

[P4] 纵向联邦学习建模优化方法、设备、介质及程序产品。徐梦炜，武耀宗，**蔡栋琪**，王尚广

学术服务

- **TPC Member**

MobiSys'26, EIS'25, MobiSys'24 AE, MobiCom'24 AE, NCSC-edge'22, TURC-SIGBED-China'23

- **Conference Reviewer**

ICLR'26, NeurIPS'26, ICASSP'26, ICASSP'24, ICASSP'25, SAGC'22

- **Journal Reviewer**

ACM CSUR, PACM IMWUT, ACM TIOT, Scientific Reports, TSC, TMC, TKDE, TECS, IoTJ, PeerJ

- **External Reviewer**

MLSys'26, MLSys'25, ICWS'24, IEEE EDGE'24, IEEE EDGE'23, ICWS'23, EIS'21

教学经历

- 助教，机器学习系统原理，剑桥大学，2024

参与项目

1. 国家重点研发计划项目（科技部），面向大规模分布式人工智能应用的关键网络技术研究，2020.07-2024.01，20M，已结题，项目骨干（技术研究、系统集成开发、验收结项）
2. 国家重点研发计划项目（科技部），跨域异质分布式学习和推理系统，2021.08-2024.12，75M，已结题，项目骨干（项目申报、技术研究、系统集成开发、验收结项）

3. 校企合作（小米集团），端侧大模型的个性化高效微调关键技术研究，2024.09–2025.09，0.18M，在研，项目骨干（项目申报、技术研究）
4. 创新基金（北京邮电大学），面向复杂自然语言模型的联邦小样本学习方法研究，2023.4-2024.04，0.012M，已结题，项目负责人（独立 PI）
5. 校企合作（微众银行），可信联邦学习算法研究及应用 - 可信联邦大模型研究，2023.09-2024.09，0.2M，已结题，项目骨干（项目申报、技术研究、系统集成开发、验收结项）

受邀汇报/讲座

- EMDL'21 (Co-located with MobiSys'21), Towards ubiquitous learning: A first measurement of on-device training performance, Online, 2021/06/25
- EuroMLSys'23 (Co-located with EuroSys'23), Towards Practical Few-shot Federated NLP Rome, Italy, 2023/05/08
- MobiCom'23, Efficient Federated Learning for Modern NLP, Madrid, Spain, 2023/10/05
- MobiCom'23, Federated Few-shot Learning for Mobile NLP, Madrid, Spain, 2023/10/05
- Northwestern Polytechnical University, PhD Research Methodology, Online, 2023/10/30
- BUPT 'Diligent Research, Academic Leadership' Academic Forum, Efficient Federated Learning for Modern NLP, Beijing, China, 2023/12/26
- EuroMLSys'24 (Co-located with EuroSys'24), FedRDMA: Communication-Efficient Cross-Silo Federated LLM via Chunked RDMA Transmission, Athens, Greece, 2024/04/22
- MoiSys'24 N2Women, Large Language Models on Mobile Devices: Measurements, Analysis, and Insights, Tokyo, Japan, 2024/06/03
- EdgeFM'24 (Co-located with MobiSys'24), Large Language Models on Mobile Devices: Measurements, Analysis, and Insights, Tokyo, Japan, 2024/06/07
- USENIX ATC'24, FwdLLM: Efficient Federated Finetuning of Large Language Models with Perturbed Inferences, SANTA CLARA, CA, USA, 2024/07/11
- AI TIME NeurIPS 2024 Forum, SILENCE: Protecting Privacy in Offloaded Speech Understanding on Resource-constrained Devices, Online, 2024/11/20
- NeurIPS'24, SILENCE: Protecting Privacy in Offloaded Speech Understanding on Resource-constrained Devices, Vancouver, Canada, 2024/12/11
- CCF Talk, Efficient Federated Learning System for LLMs, Online, 2024/12/22
- Cambridge ML Systems Seminar Series, Training LLMs Anywhere: Enabling Large-Scale Decentralized Learning on Your Mobiles Devices, Cambridge, UK, 2025/1/28
- Department of Computer Science and Technology, Efficient Machine Learning System for Mobile Devices, Soochow University, China, 2025/04/22